

Solid Reflexivity Of The Phantom

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Summary

Pass 78 tests whether the all-prime phantom

$$\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$$

is reflexive in the derived/solid target. The answer is positive, but the reflexivity carries an antipode sign coming from the single odd shift of the phantom.

Main Claims

- **Single dual.** The dual of ϵ is represented by a degree-one extension object; finite shadows give $\mathrm{Hom}(\mathbb{Z}/N, \mathbb{Z}) = 0$ and $\mathrm{Ext}^1(\mathbb{Z}/N, \mathbb{Z}) \cong \mathbb{Z}/N$.
- **Double dual.** The unit class $c = 1$ gives an isomorphism at every checked finite stage, so $D^2(\epsilon) = \epsilon$. A pathological idempotent class creates a secondary phantom, showing that the unit-class hypothesis is doing real work.
- **Antipode sign.** The shadow has $D^2 = +\mathrm{id}$, while the degree-one phantom shift contributes the Koszul sign -1 .

Machine Verification

`code/scripts/check-pass78.py` produced `artifacts/reports/pass78-solid-reflexivity-phantom-check.json` with overall PASS. It verifies the degree-one dual, double-dual recovery through the unit class, the secondary-phantom control example, and the sign calculation.

Next Use

Pass 78 supplies the distinction used later: ϵ is reflexive, but this does not imply tensor dualizability or the existence of a Fourier/Weyl flip.